Eliza Hahn Neights

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Education

2021-2027	**George Washington University**	Ph.D in Physics (*in progress*) *Advisors: Dr. Sylvain Guiriec & Dr. Carolyn Kierans*
2021-2024	**George Washington University**	M.Phil in Physics M.S. in Physics
2017-2021	**Pennsylvania State University**	B.S. in Astronomy & Astrophysics B.S. in Physics Minor in Mathematics

Publications

In addition to the publications below, I have led 12 GCN circulars and 1 Astronomer's Telegram.
E. Neights et al. 2026 was featured as a in *Nature Astronomy* Research Highlight in Feb 2026.
Orcid: 0009-0005-0762-4507, SciX

Refereed

E. Neights, E. Burns, C. L. Fryer, *et al.*, "GRB 250702B: discovery of a gamma-ray burst from a black hole falling into a star," *MNRAS*, Jan. 2026. DOI: 10.1093/mnras/staf2019.

N. Parmiggiani, A. Bulgarelli, G. Panebianco, *et al.*, "COSI Short Gamma-Ray Burst Localization Using BGO Shield Data," *ApJ*, Feb. 2026. DOI: 10.3847/1538-4357/ae25fc.

A. Rizzo, N. Parmiggiani, A. Bulgarelli, *et al.*, "Comparing Classical and Quantum Deep Learning Techniques for Anomaly Detection of Short-Duration Gamma-Ray Signals," *Astronomy and Computing*, Jul. 2026. DOI: 10.1016/j.ascom.2026.101090.

A. C. Trigg, E. Burns, M. Negro, *et al.*, "From Rare Events to a Population: Discovering Overlooked Extragalactic Magnetar Giant Flare Candidates in Archival Fermi Gamma-ray Burst Monitor Data," *arXiv e-prints*, Oct. 2025. DOI: 10.48550/arXiv.2510.23367.

H. Yoneda, T. Siegert, I. Martinez-Castellanos, *et al.*, "Enhancing Compton telescope imaging with maximum a posteriori estimation: A modified Richardson–Lucy algorithm for the Compton Spectrometer and Imager," *A&A*, May 2025. DOI: 10.1051/0004-6361/202453528.

H. C. Gulick, **E. Neights**, S. Al Nussirat, *et al.*, "Across the soft gamma-ray regime: utilizing simultaneous detections in the Compton Spectrometer and Imager (COSI) and the Background and Transient Observer (BTO) to understand astrophysical transients," in *Space Telescopes and Instrumentation 2024: Ultraviolet to Gamma Ray*, Aug. 2024. DOI: 10.1117/12.3020606.

Non-Refereed

E. Burns, J. Andrews, R. Szabo, *et al.*, "The Heavy Element Enrichment History of the Universe from Neutron Star Mergers with Habitable Worlds Observatory," *arXiv e-prints*, Jul. 2025. DOI: 10.48550/arXiv.2507.09778.

E. Burns, C. L. Fryer, I. Agullo, *et al.*, "Multidisciplinary Science in the Multimessenger Era," *arXiv e-prints*, Feb. 2025. DOI: 10.48550/arXiv.2502.03577.

H. Yoneda, T. Siegert, I. Martinez-Castellanos, *et al.*, "Imaging MeV Gamma-ray Lines with Advanced Image Reconstruction Framework for COSI," in *39th International Cosmic Ray Conference*, Dec. 2025. URL: https://pos.sissa.it/501/891/pdf.

J. Tomsick, S. Boggs, A. Zoglauer, *et al.*, "The Compton Spectrometer and Imager," in *38th International Cosmic Ray Conference*, Sep. 2024. DOI: 10.22323/1.444.0745.

I. Martinez-Castellanos, S. Gallego, C.-Y. Huang, *et al.*, "The cosipy library: COSI's high-level analysis software," *arXiv e-prints*, Aug. 2023. DOI: 10.48550/arXiv.2308.11436.

T. Akindele, T. Anderson, E. Anderssen, *et al.*, "A Call to Arms Control: Synergies between Nonproliferation Applications of Neutrino Detectors and Large-Scale Fundamental Neutrino Physics Experiments," *arXiv e-prints*, Feb. 2022. DOI: 10.48550/arXiv.2203.00042.

T. Grégoire, H. Ayala Solares, S. Coutu, *et al.*, "Model independent search for transient multimessenger events with AMON using outlier detection methods," in *37th International Cosmic Ray Conference*, Mar. 2022, p. 934. DOI: 10.22323/1.395.0934.

Awards & Scholarships

2025 **Gus W. Weiss Prize (Honorable Mention)**
Award celebrating graduate student research contributions and dedication to George Washington University Physics department and larger physics community.

2022-2025 **John Mather Nobel Scholar**
Travel allowances awarded by John and Jane Mather Foundation for Science and the Arts towards cost of presenting research at conferences.

2017-2021 **Braddock Scholarship**
Four-year academic merit scholarship awarded by Penn State Eberly College of Science.

2019-2021 **M. Dean and Jean L. Underwood Scholarship in Physics**
Academic merit scholarship awarded by Penn State Eberly College of Science.

2019-2020 **Eberly College of Science Undergraduate Research Award**
Funding from Office of Science Engagement in Penn State Eberly College of Science to do undergraduate astrophysics research with Dr. Doug Cowen.

2017-2020 **Dean's List**
Maintained GPA above 3.5.

2018 **NASA Pennsylvania Space Grant**
Women in Science and Engineering Research scholarship to do undergraduate astrophysics research with Dr. Miguel Mostafá.

Presentations

Invited Talks

Mar 2026 **Discovery of a Gamma-Ray Burst from a Black Hole Falling into a Star** American Physical Society Global Physics Summit (*Denver, CO*)

Oct 2025 **COSI GRB Group Update** *COSI* Collaboration Meeting (*St. Louis, MO*)

Oct 2024 **COSI's Spectral Analysis Software** *COSI* Collaboration Meeting (*Berkeley, CA*)

Oct 2024 **GRB Polarization with COSI** *COSI* Collaboration Meeting (*Berkeley, CA*)

Apr 2024 **The COSI GRB Science Goals** "Exploring the MeV Gamma-ray Sky: The Past, Present, and Future" Special Session at the High Energy Astrophysics Division Meeting (*Horseshoe Bay, TX*)

Jul 2024 **GRB Polarization with COSI** GRB Forum (*Athens, Greece*)

Contributed Talks

Oct 2025 **GRB Localizations and Polarimetry with COSI** "Exploring the MeV Gamma-ray Sky with *COSI*" Special Session at the High Energy Astrophysics Division Meeting (*St. Louis, MO*)

Presentations (continued)

Jan 2025	**Gamma-Ray Burst Polarization with COSI** American Astronomical Society Meeting (*National Harbor, MD*)
Aug 2024	**Studying GRBs Using COSI's Polarization Measurements** AstroCon DC (*Washington, DC*)
Jun 2024	**Studying GRBs Using COSI** *Fermi* Summer School (*Lewes, DE*)
May 2024	**GRB Science Using COSI** Seminar at Oregon State University (*Corvallis, OR*)
Apr 2024	**Studying Gamma-Ray Bursts Using COSI** High Energy Astrophysics Division Meeting (*Horseshoe Bay, TX*)
Aug 2023	**Using COSI to Study GRBs** DMV Consortium Graduate Student Astrophysics Conference (*Washington, DC*)
Aug 2022	**Scientific Pipeline Development for COSI** NASA Goddard Space Flight Center Astrophysics Summer Intern Symposium (*Virtual*)
Sep 2020	**Archival Coincidence Analysis for IceCube Cascade Events & GRBs** IceCube Collaboration Meeting (*Virtual*)

Posters

Oct 2025	**Gamma-Ray Observations of GRB 250702B: Gamma-Ray Burst from a Black Hole Falling into a Star** High Energy Astrophysics Division Meeting (*St. Louis, MO*)
Mar 2023	**Introduction to COSI Data Analysis** High Energy Astrophysics Division Meeting (*Waikoloa, HI*)
May 2020	**Photomultiplier Tube Pressure Testing for WATCHMAN** Pennsylvania State University Undergraduate Exhibition (*State College, PA*)
Nov 2018	**The Study of Binary Neutron Star Mergers from Simulations** Pennsylvania State University Undergraduate Research Symposium (*State College, PA*)

Selected Press

2026 **BBC Sky at Night**
`skyatnightmagazine.com/news/grb-250702b-study-interview`

2025 **National Geographic**
`nationalgeographic.com/science/article/gamma-ray-burst-black-hole-eats-star`

NASA
`science.nasa.gov/science-research/black-hole-eats-star`

Sky and Telescope
`skyandtelescope.org/astronomy-news/black-hole-eats-through-star-explodes-it-from-within`

GW Today
`gwtoday.gwu.edu/black-hole-eats-star-student-charts-record-blast`

Leadership & Teaching

2024-present **Burst Advocate for Fermi-GBM**
Volunteer for 6-8 burst advocate shifts per month for *Fermi* Gamma-ray Burst Monitor. Analyze triggered observations, ensure good data quality, and send alerts to scientific community.

2023-present **Technical Lead for COSI GRB Science Team**
Plan and lead monthly meetings for >50 person group. Manage GRB team tasks, including development of *COSI* transient analysis pipelines. Lead creation of instrument simulations of *COSI* transient observations which are used to develop analysis software, test onboard transient trigger algorithm, and make GRB science predictions. Lead development of population-level simulation and analysis software for *COSI* transient observations.

Leadership & Teaching (continued)

2022-present **Spectral and Polarization Analysis Software Development for COSI**
Lead development of publicly available spectral and polarization analysis software used by scientific community to analyze *COSI* data.

2021-2022 **Graduate Teaching Assistant**
Assisted in teaching and graded for introductory undergraduate physics classes of 25-50 students.

Outreach & Professional Development

2025 **Attendee of Summer School in Statistics for Astronomers**

2024 **Attendee of Fermi Summer School**

Volunteer for Astronomy Festival on the National Mall

2023-2024 **Member of COSI Collaborations Connections Team** Addressed issues of diversity, equity, inclusion, and accessibility in *COSI* collaboration.

2020 **Volunteer for Penn State AstroFest**

2019-2021 **Safety Liaison for Penn State Social Dance Club**
Wrote safety and inclusivity policies for club addressed concerns of club members, and handled cases of misconduct.

2018 **Presenter at Penn State AstroFest**

Collaborations

COSI
Technical lead for GRB science team, leading creation of transient observation simulations, making GRB science predictions, and leading development of spectral and polarization analysis software.

Fermi-GBM
Burst advocate and led analysis of record-duration GRB 250702B.